A Fixed Confidence Gate for Selective Classification

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Abstract

We examine a fixed confidence gate for selective classification using a synthetic evaluation fixture. The gate abstains from predictions below a specified confidence threshold. We report accuracy among accepted predictions alongside coverage on two synthetic datasets. Conditional accuracy increases on one dataset and decreases on the other; these observations do not establish an overall accuracy improvement. A range bound for selective risk requires positive coverage and bounded loss. Statistical uncertainty, runtime, and distribution shift remain unevaluated.

1. Introduction

Selective classification allows a model to abstain from uncertain predictions. We study a fixed confidence gate, rather than a learned rejection policy. This synthetic evaluation asks whether a fixed threshold changes accuracy among accepted predictions. Coverage is reported alongside conditional accuracy because the gate changes which predictions are evaluated.

2. Method

Gate uses a fixed classifier and accepts a prediction when its maximum class probability is at least 0.8; otherwise, it abstains. Always uses the same classifier and accepts every prediction. Neither method retrains the model. For a fixed classifier h and acceptance function $a(x) \in \{0, 1\}$, assume $0 \leq \ell(h(x), y) \leq 1$ and positive coverage $C = \mathbb{E}[a(X)] > 0$. Selective risk is

$$R_{\text{sel}} = \frac{\mathbb{E}[a(X)\ell(h(X), Y)]}{C}. \tag{1}$$

Theorem 1 *Under these assumptions,* $0 \leq R_{\text{sel}} \leq 1$.

The result is a range bound, not a generalization guarantee or a comparison between Gate and Always. Appendix A gives the full proof.

3. Synthetic Evaluation

Alpha and Beta each contain 1000 synthetic samples. Both methods use the same model and 5 evaluation seeds; results are seed means, with no variance estimates. Table 1 shows a higher conditional accuracy on Alpha and a lower conditional accuracy on Beta. Different evaluation populations prevent interpreting these differences as an overall accuracy improvement. No significance tests, runtime measurements, or distribution-shift evaluations were performed.

Table 1. Synthetic fixture values. Always accuracy uses all samples; Gate accuracy is conditional on acceptance. These values are not research findings.

Dataset	Always (%)	Gate (%)	Coverage (%)
Alpha	80.0	84.0	60.0
Beta	76.0	75.5	58.0

4. Limitations and Conclusion

A single baseline and synthetic datasets do not establish state-of-the-art performance. The classifier and data-generation details are **AUTHOR INPUT PENDING:** required for reproducibility. The fixed threshold does not guarantee lower risk, and deployment benefits remain untested.

Impact Statement

AUTHOR INPUT PENDING: Describe the actual impacts without making unsupported assurances.

Related Work

AUTHOR INPUT PENDING: Supply verified nearest references; no research citation has been invented.

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A. Proof of the Range Bound

Since $0 \leq a(X)\ell(h(X), Y) \leq a(X)$, taking expectations gives $0 \leq \mathbb{E}[a(X)\ell(h(X), Y)] \leq C$. Division by $C > 0$ proves Theorem 1. This argument applies to the specified distribution; it does not establish a bound under distribution shift.